

Classifying rejection from kidney biopsy RNA

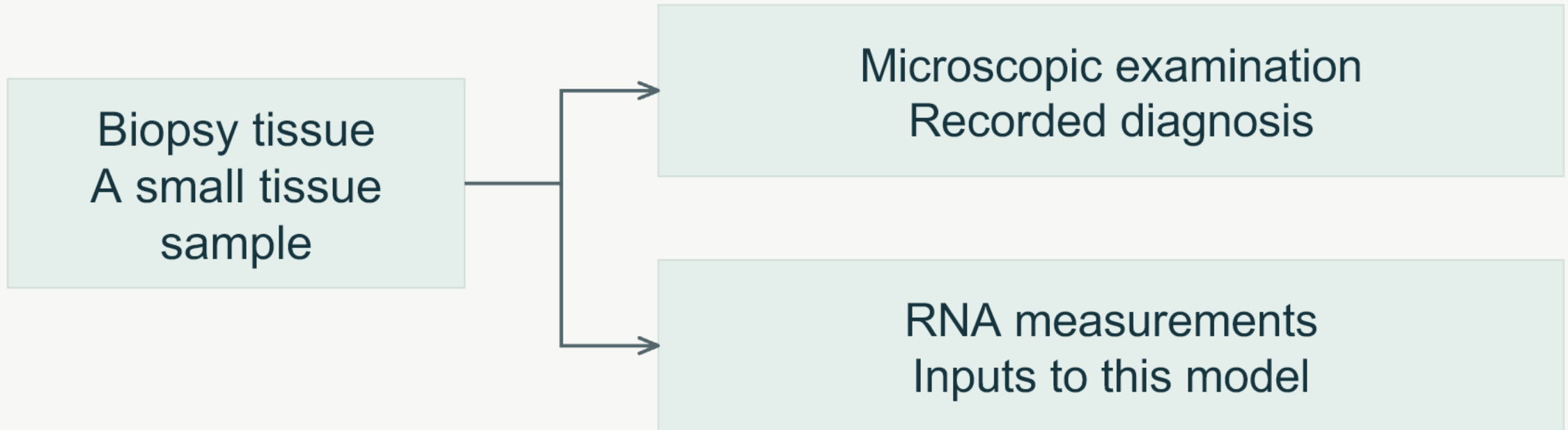
Can molecular measurements classify
the recorded biopsy diagnosis?

25 missed rejection cases and 8 false flags

CatBoost on 345 author technical-validation specimens

Kidney biopsy and transplant rejection

Rejection: the recipient's immune system attacks the donated kidney.

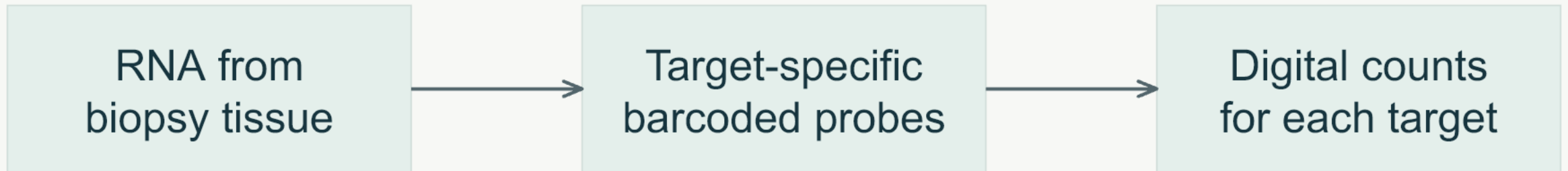


The prediction concerns this existing specimen.

An assay measures specified molecules

RNA is made as genes are expressed.

The assay counts signals for selected RNA targets.



770 measured targets

758 model inputs + 12 housekeeping targets

One row describes one biopsy specimen

1,395 archived kidney transplant biopsies

Recorded diagnosis	Binary label
No rejection	0
Antibody-mediated rejection	1
T-cell-mediated rejection	1
Mixed rejection	1

Inputs: assay counts Reference label: recorded diagnosis

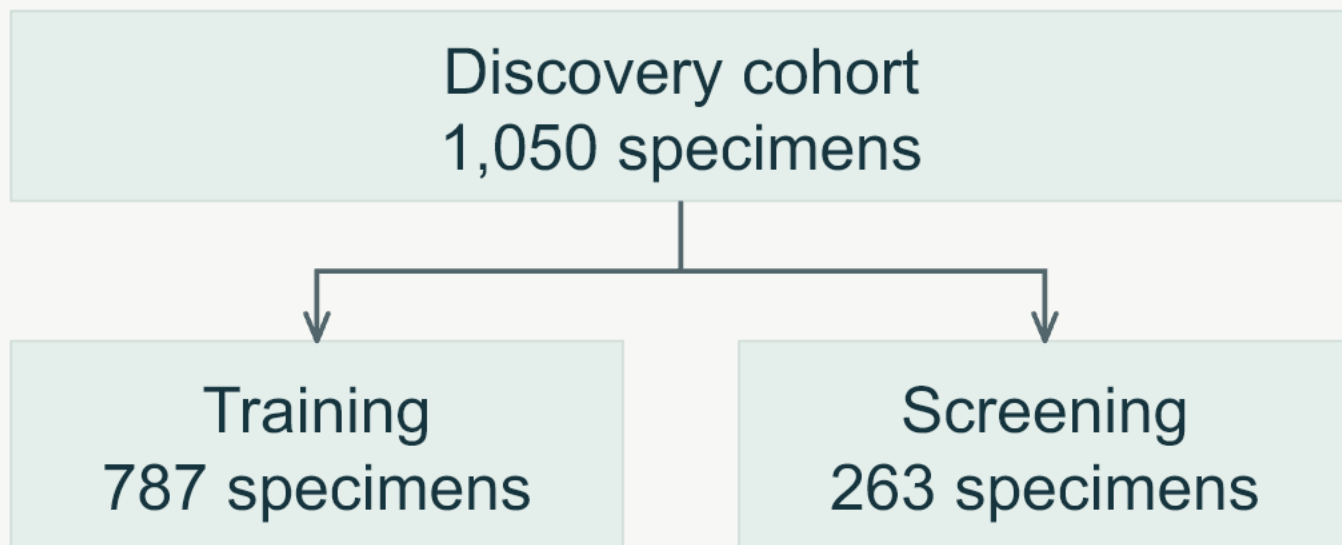
Normalization uses each specimen's reference targets

Housekeeping targets provide a within-specimen reference.

IFNG in this specimen	Value
Raw count	4
$\log_2(\text{count} + 1)$	2.322
Mean of 12 housekeeping log counts	7.900
Normalized input: $2.322 - 7.900$	-5.578

Apply to all 758 inputs. Fit later learned steps on training data.

Model choice and thresholds use discovery data



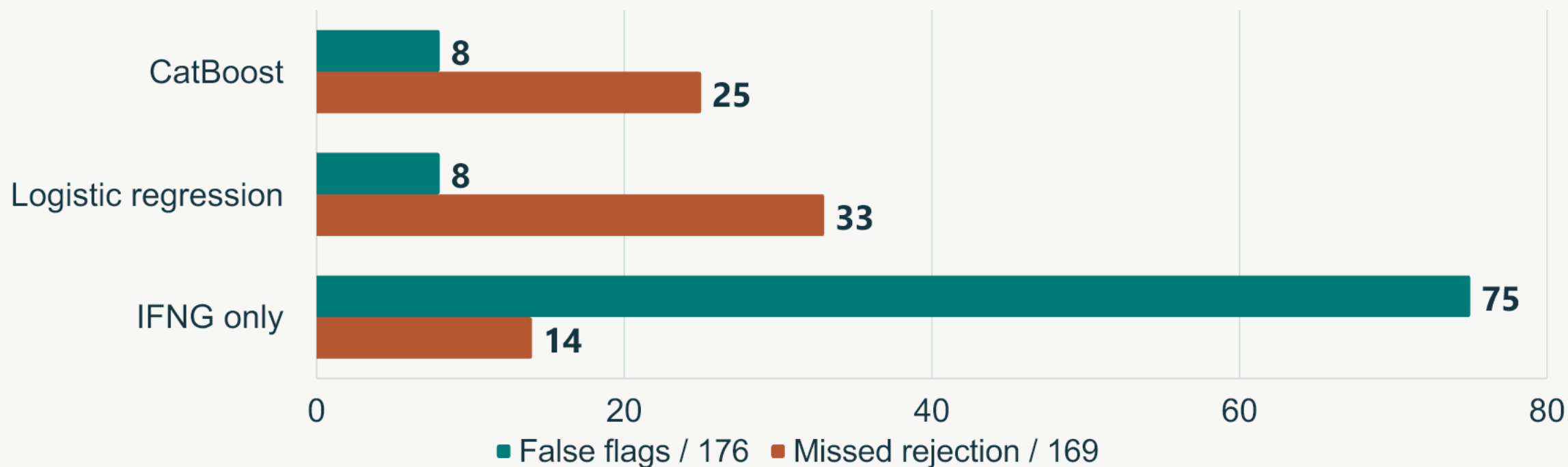
Author technical validation
345 specimens
Evaluate frozen choices

Screening rule

Retain at least 90% of rejection cases, then minimize false flags.

Several measurements reduced false flags

Same 345 specimens: 169 rejection and 176 no rejection



Constant baseline: 0 misses and 176 false flags. It flags every specimen.

Most CatBoost misses were T-cell rejection

144 / 169

rejection cases detected

85.2%

Below the 90%
screening target

Recorded rejection	Missed / total
Antibody-mediated	6 / 56
T-cell-mediated	18 / 95
Mixed	1 / 18

Errors mean disagreement with the recorded diagnosis.

The advantage over logistic regression is uncertain

8 fewer misses

Same 8 false flags. The paired miss-difference interval includes zero.

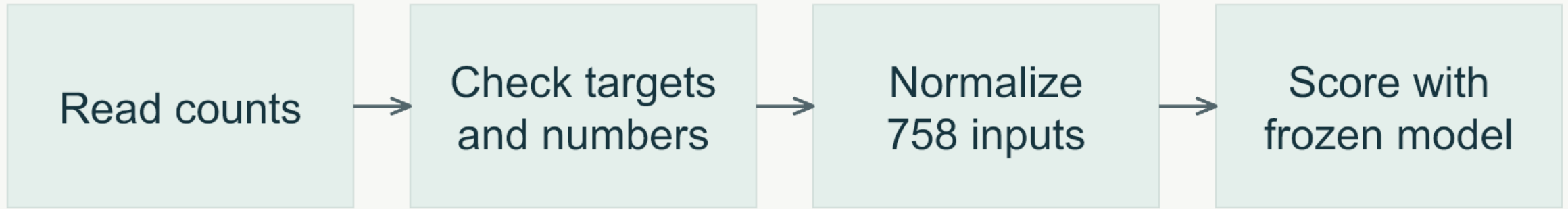
Model selected across 20 discovery splits

CatBoost	Logistic regression
13 selections	7 selections

Overlapping repetitions describe split variability.

Training and serving share the same preprocessing

CSV input



Versioned score + threshold flag

Diagnosis, specimen ID, cohort and assay batch stay out of the predictors.

One public specimen through the service

Public example: GSM6510425

Model score	Frozen threshold	Result
0.274934	0.876588	Below threshold

Remove IFNG from the same file

“Missing required assay targets: IFNG”

Model: 20260915_shared:any_rejection:catboost_all_depth4

The service reproduces all 345 saved scores

345 / 345

API, command line and saved evaluation agree within a tolerance of 10^{-12} .

94 tests passed

Local research container checked

Missing targets stop scoring. Incompatible model metadata stops loading.

The result applies to one deposited study

Established

Agreement with recorded diagnoses
for existing transplant biopsies

Next evidence needed

An independent cohort with known patient and center IDs,
and documented laboratory quality checks

Molecular counts can classify recorded rejection

25 / 169 misses. 8 / 176 false flags.

Logistic regression remains a credible simpler alternative.

The same preprocessing powers a tested research service.

Backup: complete primary comparison

Model	Misses	False flags	Sensitivity	Specificity	ROC-AUC
CatBoost	25	8	85.2%	95.5%	0.969
Logistic	33	8	80.5%	95.5%	0.956
IFNG only	14	75	91.7%	57.4%	0.842
Constant	0	176	100.0%	0.0%	0.500

Denominators: 169 rejection; 176 no rejection. Fixed thresholds per model.

Backup: normalization and threshold selection

Normalized target = $\log_2(\text{raw count} + 1)$
– mean of the 12 housekeeping $\log_2(\text{count} + 1)$ values

Logistic: scaling fitted on training rows; regularization $C = 0.1$.

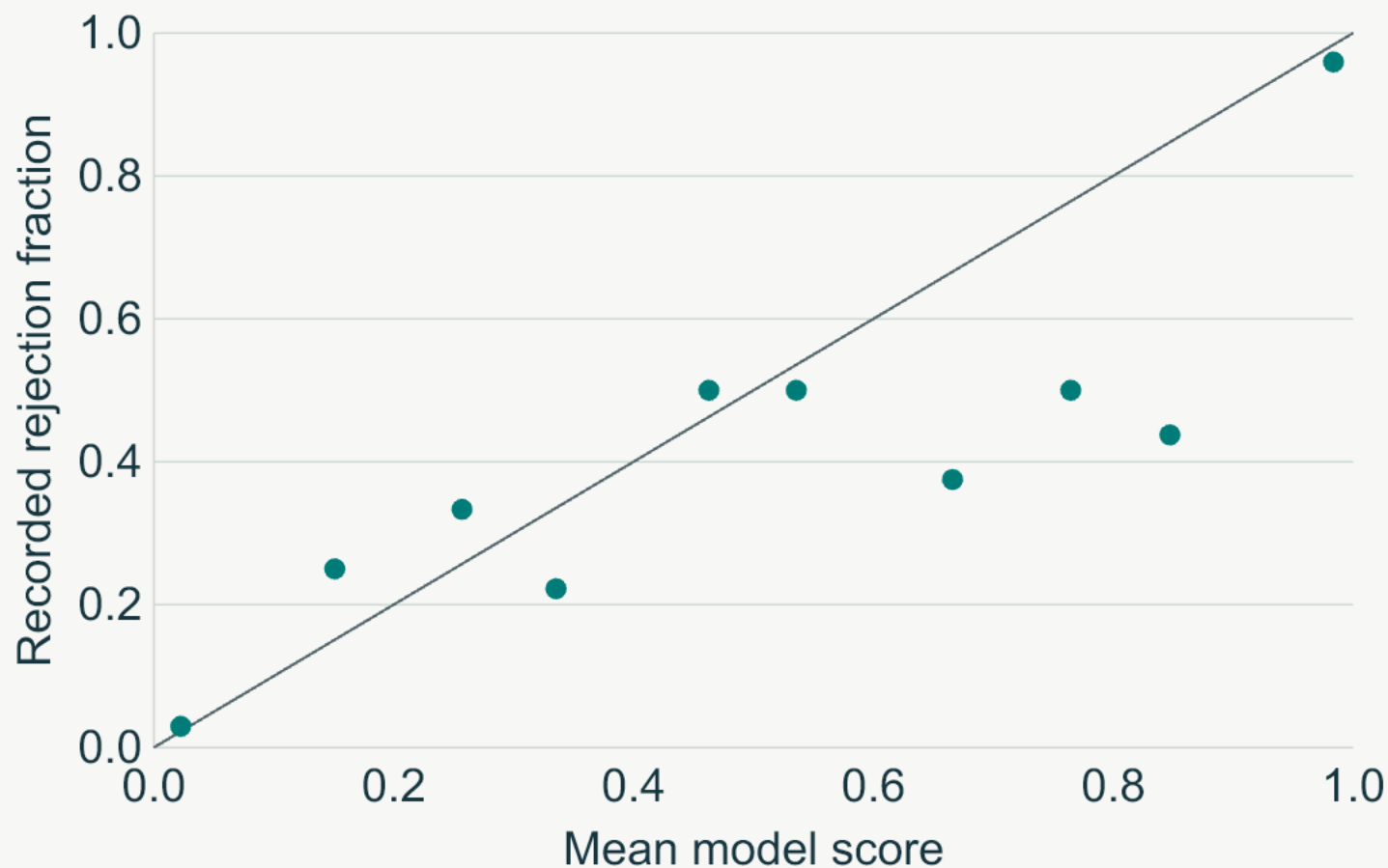
CatBoost: 300 trees, depth 4, learning rate 0.04.

Both detected 157 / 174 screening rejection cases.

CatBoost made 5 false flags; logistic made 6.

Flag when score ≥ 0.8765880870219778 for the frozen service.

Backup: ranking does not guarantee calibrated scores



Score bin 0.8–0.9

0.847

mean score

7 / 16

had recorded rejection
43.8% (95% CI 23.1–66.8%)

Backup: subtype follow-up retained the binary service

Model	Any-rejection misses	False flags
Binary CatBoost	25 / 169	8 / 176
Four-class CatBoost	29 / 169	8 / 176
Four-class logistic	28 / 169	10 / 176
Separate component models	24 / 169	15 / 176

Four-class logistic recognized 10 / 18 mixed diagnoses, with 5 false mixed calls.

Backup: study, assay and deployment limits

Question	What the project establishes
Independent patients / centers?	Independence not verified
Laboratory assay quality?	File and numeric checks; assay QC not repeated
Individual probabilities?	Calibration remains limited
Cloud deployment?	Local container tested; cloud work deferred

Biopsy-level resampling may understate uncertainty if specimens are related.

Backup: sources and contributions

Public study

Zhang et al., Laboratory Investigation (2024)
GSE212160 raw NanoString B-HOT measurements

This project

Binary comparison and runs produced in this repository,
shared software and a tested local demonstration

Presentation approach

Bourne (2007); Kosslyn et al. (2012); Garner & Alley (2013); Rougier et al. (2014)